# FINITENESS OF THE FIBRES OF THE POINT MAPS OF A GROUPOID PRESENTATION

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

*Editorial note.* Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

*Source reference.* Original lemma 03H4: sources/stacks-chow/tags.txt, source lines 4553–4553. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 02GL: sources/stacks-chow/tags.txt, source lines 3232–3232. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *Let $S$ be a scheme. Let $X$ be an algebraic space over $S$. Let $U$ be a scheme. Let $\varphi : U \to X$ be an étale morphism such that the projections $R = U \times_X U \to U$ are quasi-compact; for example if $\varphi$ is quasi-compact. Then the fibres of*

$$|U| \to |X| \quad \textit{and} \quad |R| \to |X|$$

*are finite.*

*Proof.* Denote $R = U \times_X U$, and $s, t : R \to U$ the projections. Let $u \in U$ be a point, and let $x \in |X|$ be its image. The fibre of $|U| \to |X|$ over $x$ is equal to $s(t^{-1}(\{u\}))$ by Lemma 03H4, and the fibre of $|R| \to |X|$ over $x$ is $t^{-1}(s(t^{-1}(\{u\})))$. Since $t : R \to U$ is étale and quasi-compact, it has finite fibres (as its fibres are disjoint unions of spectra of fields by Morphisms, Lemma 02GL and quasi-compact). Hence we win. □

*Date*: stacks-project:spaces-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.